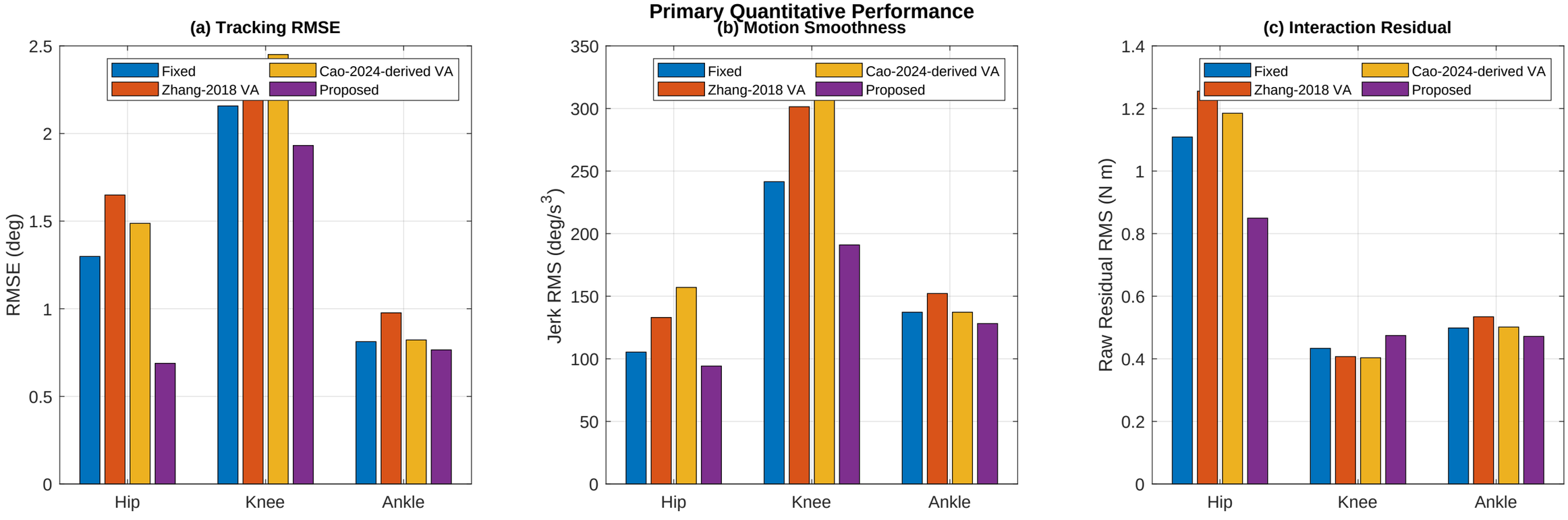
Primary Quantitative Performance
(a) Tracking RMSE
RMSE (deg)
2.5
2
1.5
1
0.5
0
Fixed
Zhang-2018 VA
Cao-2024-derived VA
Proposed
Hip
Knee
Ankle
(b) Motion Smoothness
Jerk RMS (deg/s^3)
350
300
250
200
150
100
50
0
Fixed
Zhang-2018 VA
Cao-2024-derived VA
Proposed
Hip
Knee
Ankle
(c) Interaction Residual
Raw Residual RMS (N m)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Fixed
Zhang-2018 VA
Cao-2024-derived VA
Proposed
Hip
Knee
Ankle